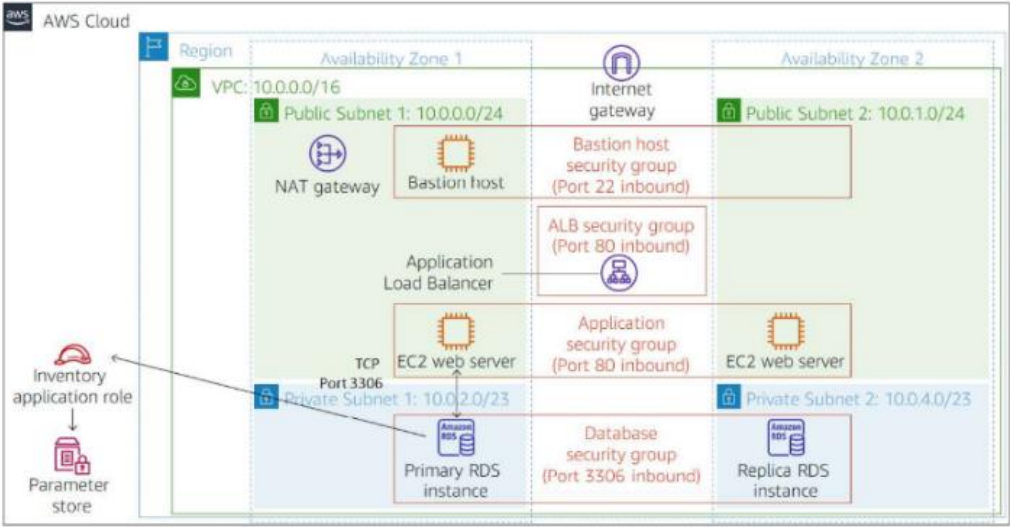


AWS CAPSTONE PROJECT

DIAGRAM:



SUMMARY:

➤ *The database:*

Under the current configuration, the web server and database are hosted on the same EC2 instance within a public subnet.

➤ *Use an Amazon RDS instance:*

Amazon RDS provides managed database services with features such as automated backups, scaling, and security patching. This offloads numerous administrative responsibilities and provides a higher level of security than manually managing a database on an EC2 server. Utilize Secrets Manager. Storing database credentials in Secrets Manager provides an additional layer of security. This guarantees that sensitive information, such as passwords, is securely handled and accessed only as needed, lowering the risk of exposure in the event of a breach.

➤ *Implement a bastion host:*

The bastion host serves as a gateway to resources in the private subnet, providing an additional layer of security. Consumers first connect to the bastion host using SSH, from which they can access additional resources on the private network. This restricts direct access to sensitive resources via the internet.

➤ *Secure Load Balancer:*

A basic security practice is to configure the application load balancer's security group to allow only necessary inbound traffic (in this case, HTTP on port 80). This ensures that only authentic traffic is routed to the application servers behind the load balancer.

➤ *NAT Gateway and internet gateway:*

Using a NAT Gateway for outbound traffic from resources on the private subnet and an internet gateway for resources on the public subnet ensures proper network connectivity while maintaining security boundaries.

➤ *Inventory Application Role and Parameter Store:*

Leveraging AWS Identity and Access Management (IAM) roles to grant rights to resources such as the inventory application together with AWS Systems Manager Parameter Store to securely store configuration data, improves the system's security and manageability.